

Comparative Study: 8086 to Pentium 4 MPCA Module 6

> **Definition:** The evolution of the **Intel x86 Microprocessor Architecture** from the 16-bit 8086 to the 7th-generation 32/64-bit Pentium 4 demonstrates radical advancements in bus widths, pipeline depths, caching hierarchies, and parallel instruction execution.

1. Comprehensive Master Comparison Table

| Feature / Processor | Intel 8086 | Intel 80386DX | Intel Pentium (P5) | Intel Pentium II | Intel Pentium 4 |

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2. Key Architectural Milestones

1. **8086:** Introduced 16-bit computing and segmented memory architecture.

2. **80**

3. Quick Recall Flow

8086
Hyper-